

Semester Mini Project - Task 3

Image Enhancement in the Spatial Domain

Course: Digital Image Processing

Project: SmartScan AI - Intelligent Document Scanner

Team Member	Contribution
Jainam Jain	Core pipeline, boundary detection, GUI, OCR
Dhanush	Dataset collection (148 photos)
Vivek	Dataset collection (104 photos)

1. Aim

To implement and analyse spatial-domain image enhancement techniques - intensity transformations, histogram operations, and image arithmetic - and integrate them as pre-processing capabilities in the SmartScan AI document scanner. These techniques serve as the foundation for the segmentation module that follows.

2. Intensity Transformation Techniques

Four classical intensity transformations were implemented and tested on document images from our self-captured dataset:

Technique	Formula	Purpose
Image Negative	$s = 255 - r$	Invert intensity; enhance bright details in dark regions
Log Transform	$s = c * \log(1 + r)$	Compress highlights, expand shadows; good for dark images
Gamma (Power-Law)	$s = c * r^{\gamma}$	$\gamma < 1$ brightens, $\gamma > 1$ darkens; exposure correction
Contrast Stretching	Piecewise linear	Stretch a narrow band to full [0,255] range

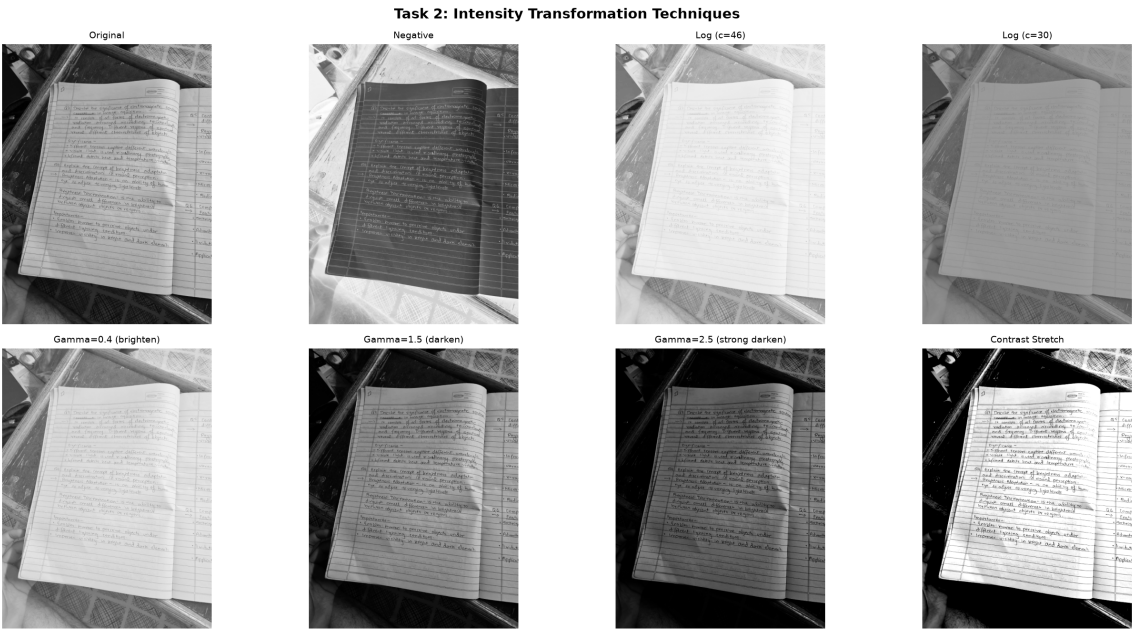


Figure 1: All intensity transformations applied to jainam_doc_02.jpg. Note how Gamma=0.4 brightens the document while Gamma=2.5 darkens it. The log transform compresses highlights significantly.

3. Histogram Analysis

Histograms visualise the distribution of pixel intensities in an image. A narrow, peaked histogram indicates low contrast, while a broad spread indicates high contrast. By comparing histograms before and after enhancement, we can quantitatively verify that a technique actually improved the image rather than relying on subjective visual judgement.

Task 3: Histogram Analysis - Before vs After Enhancement (Gamma=0.5)

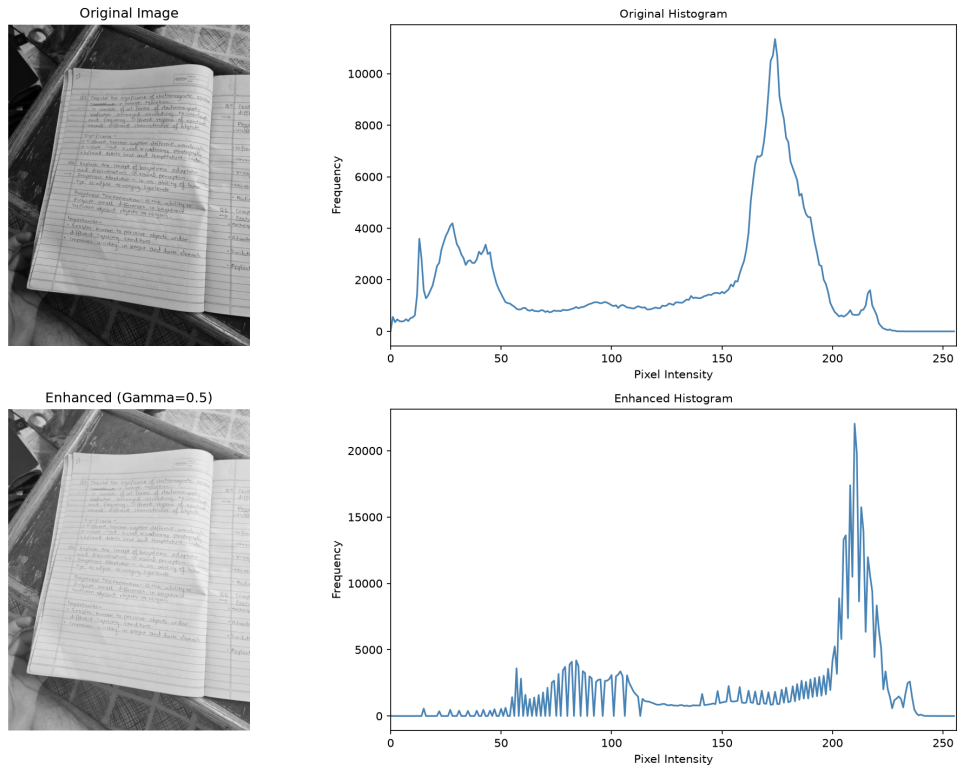


Figure 2: Histogram comparison before and after Gamma=0.5 enhancement. The original histogram is clustered in the mid-range; after enhancement, the distribution shifts rightward (brighter) and broadens.

4. Histogram Equalization

Histogram equalization transforms pixel intensities so that the output histogram is approximately uniform (flat). This maximises image contrast by spreading intensity values across the full [0, 255] range.

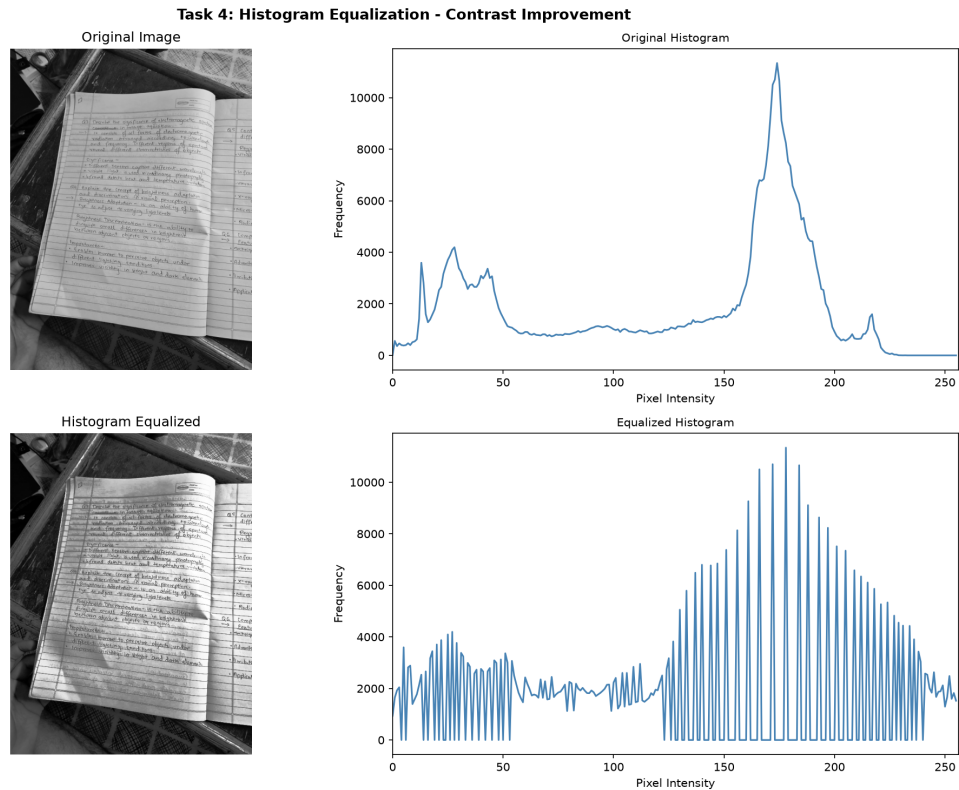


Figure 3: Original vs histogram-equalized image with their histograms. Standard deviation increased from 63.5 to 74.0, confirming improved contrast.

Observation

The original image's histogram is clustered in a narrow intensity band (low contrast). After equalization, the histogram spreads across the full [0, 255] range (std dev 63.5 -> 74.0), improving contrast especially in regions that were previously indistinguishable. However, histogram equalization can amplify sensor noise in already-bright regions, which is why our main pipeline uses CLAHE (Contrast Limited Adaptive Histogram Equalization) instead - it applies equalization locally per tile, avoiding global over-amplification.

5. Image Arithmetic Operations

Operation	Formula	Practical Application
Addition	$\alpha * I_1 + (1 - \alpha) * I_2$	Image fusion - combining multi-sensor data (e.g., visible + infrared)
Subtraction	$ I_1 - I_2 $	Change/motion detection - subtracting a reference frame reveals moving objects
Averaging	$(1/N) * \sum(I_k)$	Noise reduction - averaging N noisy captures reduces noise by $\sqrt{N}$

Task 5: Image Arithmetic Operations

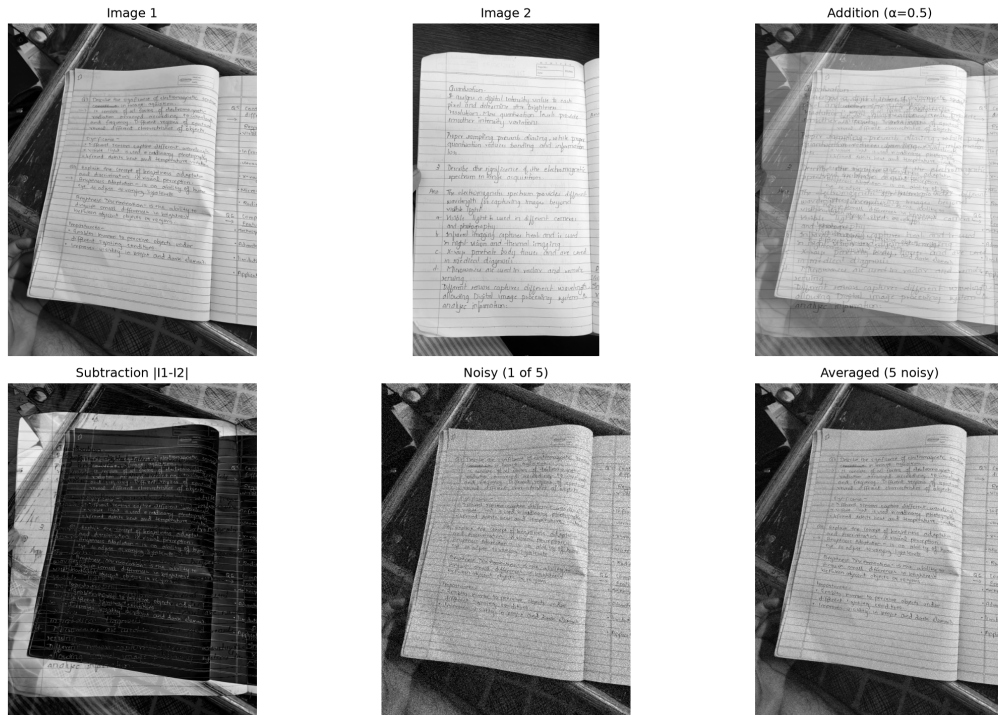


Figure 4: Top row - two source images and their weighted addition. Bottom row - absolute subtraction highlighting differences, one of five noisy copies, and the averaged result showing noise reduction.

6. Comparative Enhancement Analysis

All enhancement techniques were applied to three different images from our dataset, chosen to represent different lighting conditions and camera setups. The comparison grid below shows the visual results, and the table provides quantitative brightness (mean) and contrast (std dev) measurements.

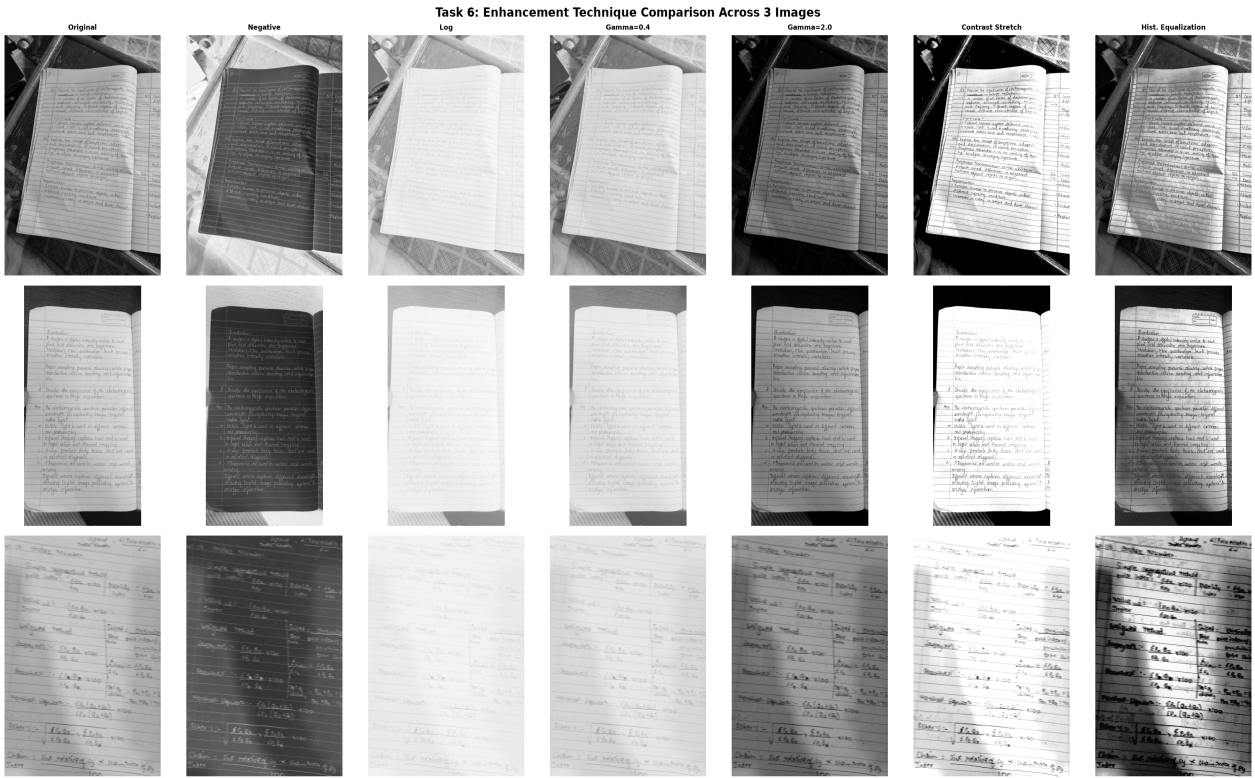


Figure 5: All 7 enhancement techniques applied across 3 different document images.

Image	Orig mean/std	Negative	Log	Gamma=0.4	Gamma=2.0	Contrast Str.	Hist. Eq.
jainam_doc_02	128/63.5	127/63.5	213/36.6	184/48.6	80/55.0	151/107.7	129/74.0
dhanush_doc_003	167/56.5	88/56.5	230/26.1	210/38.4	122/56.7	205/93.1	129/74.1
vivek_doc_010	178/19.2	77/19.2	238/5.5	220/10.1	125/25.3	234/31.6	130/74.4

Best Technique per Image

Image 1 (jainam_doc_02): Histogram Equalization - provides the best contrast spread for an indoor document with moderate lighting.

Image 2 (dhanush_doc_003): Gamma=0.4 - effectively brightens the underexposed capture without harsh redistribution artefacts.

Image 3 (vivek_doc_010): Contrast Stretching - balances detail enhancement without over-amplifying noise in an already-bright image.

7. Challenges Faced

Parameter sensitivity: Log and gamma transforms require careful tuning of c and gamma; wrong values either wash out or crush the image.

Noise amplification: Histogram equalization amplifies sensor noise alongside useful detail. This is why our main pipeline uses CLAHE (localised equalization) instead of global equalization.

Arithmetic dimension mismatch: Real dataset images have different resolutions, so arithmetic operations require explicit resizing before computation.

8. Conclusion

Spatial-domain enhancement techniques are powerful tools for improving document image quality before segmentation. Through quantitative measurement (histogram analysis, std dev comparison), we confirmed that histogram equalization and gamma correction are the most impactful for our document scanning use case. These techniques are now available as reusable modules in our project (`src/enhancement/intensity.py`, `src/enhancement/arithmetic.py`) and integrated into both the CLI and Streamlit GUI pipelines.

The key takeaway is that enhancement technique selection is image-dependent: no single transform is universally best. Histogram equalization works well for low-contrast indoor documents, gamma correction excels at exposure compensation, and contrast stretching provides controlled enhancement without noise amplification.